

Does the global SVD test in F1.1 measure what we want?

May 13, 2026

1 The question

Aggregating logits across prompts before doing SVD might flatten prompt-specific structure into a common, easier-to-fit direction. If so, the off-axis residual fraction $R(t)$ reported in F1.1 could understate the failure of per-trajectory multiplicativity, and the SVD-recovered $\bar{\beta}(k, t)$ shown in F1.2-svd could be tracking the wrong axis entirely.

This note works through what global $R(t)$ actually measures, where it can be fooled, and what the cheapest fix would be.

2 The framework claim, written per-trajectory

The imprint assumption is per-prompt:

$$\ell_t^{(n,k)} = \beta_{\text{model}}^{(n,k,t)} \cdot \ell_t^{*(n)},$$

for trajectory n , decoder index k , step t , with $\ell_t^{*(n)}$ the prompt-specific “natural” logit direction. That is the claim we want to test.

3 What F1.1 actually computes

Global $R(t)$ averages first and tests after:

$$\bar{\ell}_t(k) = \frac{1}{N} \sum_{n=1}^N \beta_{\text{model}}^{(n,k,t)} \cdot \ell_t^{*(n)}, \quad M_t = [\bar{\ell}_t(1) \mid \dots \mid \bar{\ell}_t(K)],$$

$$M_t = U \Sigma V^\top, \quad R(t) = \frac{\sum_{i \geq 2} \sigma_i^2}{\sum_i \sigma_i^2}.$$

$R(t) = 0$ means the K averaged columns lie on a one-dimensional ray. That is *necessary* for per-trajectory multiplicativity but not *sufficient*. Two failure modes hide in the gap.

4 Failure mode 1: zero-mean deviations cancel under averaging

Suppose the real per-trajectory imprint has a quadratic kicker,

$$\ell_t^{(n,k)} = \beta_k \ell_t^{*(n)} + \beta_k^2 \eta^{(n)}, \quad \eta^{(n)} \sim \text{i.i.d., zero-mean over prompts.}$$

Per-prompt, the imprint is non-multiplicative. But the aggregate kicker washes out,

$$\bar{\ell}_t(k) = \underbrace{\beta_k \cdot \frac{1}{N} \sum_n \ell_t^{*(n)}}_{\mu_t} + \underbrace{\beta_k^2 \cdot \frac{1}{N} \sum_n \eta^{(n)}}_{\rightarrow 0} \approx \beta_k \cdot \mu_t.$$

So M_t is rank-1, $R(t) \approx 0$, and the global test *certifies multiplicativity that isn't there*. This is the case the plan's html already flagged: “*Path-dependent off-axis structure that is zero-mean across trajectories cancels in the averaged test.*”

5 Failure mode 2: the principal direction picks up shared content

The deeper concern. Different prompts have different $\ell_t^{*(n)}$, but they share statistical structure — common-token directions (“the”, “of”, “and”, ...) get systematically larger logits across most natural English contexts. Call that shared direction μ_t^{avg} . When we average over N prompts,

$$\bar{\ell}_t(k) \approx \beta_k \cdot \mu_t^{\text{avg}} + (\text{prompt-specific terms, smaller in } N).$$

The SVD’s principal left singular vector u_1 lines up with μ_t^{avg} , not with any single $\ell_t^{*(n)}$. Then

$$\bar{\beta}(k, t) = \sigma_1 \cdot v_{1k}$$

measures how much each β_k -column projects onto the *average-natural* direction. That projection can track β_k closely even if per-prompt imprints are doing weirder things, as long as those weird things don’t share a direction across prompts.

This is consistent with what F1.2-svd shows. The SVD-recovered $\log \bar{\beta}$ spans only $[-1.1, -0.2]$ at $t = 199$, while the streaming $\log \hat{\beta}$ (F1.2) spans $[-4.7, +1.1]$. The SVD is finding a flatter, content-averaged direction; the streaming estimator is responding to something prompt-specific. They disagree because they are measuring different things, not because one is buggy.

6 So what does $R(t)$ measure?

The *consistent-across-prompts* off-axis structure. The observed $R(t)[199] \approx 0.155$ (F1.1) means that even after averaging out prompt-specific quirks, roughly 15% of the cross-decoder variance does not fit a single principal direction. That is a real positive finding — consistent non-multiplicativity exists.

But $R(t)$ is a *lower bound* on per-trajectory off-axis residual:

- $R(t) > 0 \Rightarrow$ consistent (across prompts) non-multiplicativity.
- $R(t) \approx 0 \Rightarrow$ consistent with per-prompt multiplicativity *or* with wildly non-multiplicative per-prompt imprints that happen to be zero-mean across prompts. Inconclusive.

7 Why per-prompt R is too noisy to break the tie

The plan anticipates this and defines a per-prompt SVD (F1.4). For each prompt p , average just that prompt’s $N = 3$ runs, form $M_t^{(p)} \in \mathbb{R}^{V \times K}$, and compute $R^{(p)}(t)$. With the imprint exactly multiplicative per-prompt, $M_t^{(p)}$ is rank-1 and $R^{(p)}(t) = 0$.

Sampling-noise floor for this estimator. Model

$$\ell_t^{(n,k)} = \beta_k \ell_t^{*(n)} + \varepsilon_t^{(n,k)}, \quad \varepsilon_t^{(n,k)} \sim \text{i.i.d., per-entry variance } \sigma^2,$$

and average over N runs at fixed prompt. The K -column matrix $M_t^{(p)}$ has signal Frobenius-squared $\|\tilde{\ell}^*\|^2 \|\beta\|^2$ and noise Frobenius-squared $VK\sigma^2/N$. SVD’s top singular value captures the signal; the remaining $K - 1$ singular values capture the orthogonal noise. Assuming $\sigma \approx \sigma_\ell$ (per-trajectory variance comparable to the natural logit magnitude, since both come from the same model on similar prefixes),

$$\mathbb{E}[R^{(p)}] \approx \frac{(K-1)/N}{\|\beta\|^2 + K/N}.$$

With $K = 6$, $N = 3$, and $\|\beta\|^2 = \sum_k \beta_k^2 \approx 10$ (for the sweep $\{0.5, 0.7, 1.0, 1.3, 1.7, 2.0\}$),

$$\mathbb{E}[R^{(p)}] \approx \frac{5/3}{10+2} \approx 0.14.$$

The observed gap between per-prompt and global in F1.4 is $\approx 0.32 - 0.16 = 0.16$, so the bulk of the per-prompt curve is finite- N noise. With $N = 3$ we cannot distinguish “consistent + per-prompt noise” from “consistent + per-prompt failure.”

For the global estimator with $N = 96$ trajectories the same calculation gives

$$\mathbb{E}[R^{\text{global}}] \approx \frac{5/96}{10+6/96} \approx 0.005,$$

so the global noise floor is $\sim 0.5\%$ and the observed $R(t)[199] \approx 0.155$ is genuine signal — but, again, only of the consistent-across-prompts component.

8 What would actually test what we want

1. **More runs per prompt.** Bumping to $N = 30$ drops the per-prompt noise floor to ≈ 0.02 and makes F1.4 the direct test. Cost: $10\times$ GPU on the existing layout.
2. **Toy validation with controlled ℓ^* .** A single deterministic-context setting on actual transformer logits, so aggregation across prompts isn’t a confound. The walkthrough’s analytic toy already does this for the corrector; a logit-level toy would close the gap for the SVD test specifically.
3. **Decompose $R(t)$ into shared vs. prompt-specific.** Project each per-prompt $u_1^{(p)}$ onto the global u_1 and look at the distribution. If the projections are all ≈ 1 , the global SVD is finding the right direction. If they spread, the global u_1 is the wrong axis and the F1.2-svd compression is a pure aggregation artifact.

9 Why F2 is mostly unaffected

F2.1 and F2.2 compare the streaming $\hat{\beta}_t$ against the SVD-recovered $\bar{\beta}_t$. The streaming estimator is computed per-trajectory and only averaged *after* taking logarithms, so it is not subject to the directional-aggregation failure mode. The gap shown in F2.1 (growing with t , anti-symmetric in $\log \beta_{\text{dec}}$) is real evidence that the model is not Bayes-optimal under the assumed prior, independent of whatever the global SVD’s $\bar{\beta}$ is doing. So even if F1.1 and F1.2-svd require caveats, F2’s headline “calibration gap grows with t ” is solid.

10 Bottom line

F1.1 is honest about *consistent-across-prompts* structure: $R(t)[199] \approx 0.155$ is genuine non-multiplicativity. But it cannot rule out larger, prompt-specific failures, and the SVD’s recovered $\bar{\beta}$ is biased toward a content-averaged direction that does not reflect any single prompt’s natural axis. The per-prompt SVD (F1.4) is the right test, but at $N = 3$ runs/prompt it is dominated by finite-sample noise. Cheapest path to a definitive answer: rerun with $N \geq 30$.